

PROJECTS

Bearings loading test rig

[Baker Hughes, a GE Company](#)

📅 05/2018 - 12/2018 📍 Florence, Italy

- Conceptual design of a novel bearings loading test rig aimed at validating bearings NTIs.
- Investigated on modeling methods of EM coupling between power (variable speed drive system) and signals cabling systems to predict potential interference issues.

GE Aviation's Catalyst

[Baker Hughes, a GE Company](#)

📅 06/2017 - 12/2018 📍 Prague, Czech Republic

- Led special instrumentation design and global technical reviews of FETT with design and testing teams (Czech Rep., Poland, US, Canada)
- Contributed to >4 M\$ cross-P&L project and customer satisfaction.

Gas Turbine embedded SAW sensors

[Baker Hughes, a GE Company](#)

📅 02/2018 - 12/2018 📍 Florence, Italy

- Co-advised MEng thesis student with University of Florence to investigate into SAW sensors high frequency numerical modeling for accuracy control of temperature measurements.

LM9000: 65 MW aeroderivative gas turbine

[Baker Hughes, a GE Company](#)

📅 06/2018 - 12/2018 📍 Florence, Italy

- Contributed to conceptual instrumentation design and test rigs erection, data acquisition architecture and control system design by technical and HAZOP reviews.

Cross'd

📅 01/2018 - 11/2018 📍 Florence, Italy

- Social networking app PoC based on Bluetooth Low Energy ranging. Made with Apache Cordova and an AWS EC2 instance

BLE Asset Tracking

📅 07/2017 - 09/2017 📍 Florence, Italy

- Sub 5\$/Unit PoC of BLE tags (TI's CC2640) with firmware optimized for long-battery life, aimed at asset tracking and inventory management.

EDUCATION

Doctor of Philosophy - PhD, Electronics Engineering

[Università degli Studi di Firenze](#)

📅 01/2011 - 02/2014 📍 Florence, Italy

School of "RF, Microwaves and Electromagnetics". Computational Methods focus. With grant (Jan 2011 - Jun 2013).

Master's Degree, Electronics Engineering

[Università degli Studi di Firenze](#)

📅 10/2007 - 04/2010 📍 Florence, Italy

Erasmus Placement/LLP framework thesis semester at the Lehrstuhl für Theoretische Elektrotechnik, Universität des Saarlandes. With grant (Sep 2009 - Mar 2010).

Bachelor's degree, Electronics Engineering

[Università degli Studi di Firenze](#)

📅 01/2003 - 04/2008 📍 Florence, Italy

Thesis project at the Laboratorio di Tecnologia per i Beni Culturali, Università degli Studi di Firenze.

High School Diploma

[École Autonome Belge à Bujumbura](#)

📅 09/1996 - 06/2002 📍 Bujumbura, Burundi

AWARDS



EMEA TAM of the Year | Ansys (01/2021)

Award for customer and industry focus



Learn and adapt to win | BHGE (07/2018)

Patent impact award for the disclosure of "Module with sensors arrangement"



Customers determine our success | BHGE (03/2017)

Bronze award for the GE Aviation's Catalyst FETT advanced test instrumentation consulting



Customers determine our success | GE (10/2016)

Bronze award for the successful execution of the vibration monitoring system ZADCO customer structural vibrations acceptance tests



Learn and adapt to win | GE (03/2016)

Bronze award for the co-generation plant cost savings on the SNC1-2 steam turbine performance validation



Learn and adapt to win | GE (10/2015)

Bronze award for the introduction and validation of an effective, cost-saving, structural vibrations digital integration algorithm aimed at accurately deriving speed & displacements measurements from ATEX-compliant accelerometers

GE Power slip-ring digital telemetry

[Baker Hughes, a GE Company](#)

📅 04/2017 - 05/2017

📍 Florence, Italy

- Ran Ansys HFSS simulations to optimize signal integrity over slip-ring connections (power supply and ethernet transmission lines)
- Achieved shrinking of the rings spacing leading to manufacturing costs reduction
- 4 weeks-FTE cross-P&L billing and customer satisfaction

Wet gas compressors at KLAB

[GE Oil & Gas](#)

📅 02/2016 - 06/2016

📍 Haugesund, Norway

- Designed and procured validation equipment for ATEX zone 1 test bench.
- Developed Ex-d antenna and low noise amplifier system for GPS synchronization on NI's PXI systems.
- Managed IEC 60079-11 simple apparatus compliance on load cells upon using extremely low voltage (1.5 V) Ex-ia Wheatstone bridge conditioners.

Power density centrifugal compressor

[GE Oil & Gas](#)

📅 03/2014 - 05/2016

📍 Florence, Italy

- Led the commissioning Fieldbus Foundation use in testing lab for more than 240 pressure transducers (0.1% uncertainty) upon identifying the proper Firmware configuration which enabled for massive and accurate pressure measurements.
- Managed capacitive vibration probes ATEX-compliance.
- Developed electrostatic numerical model to predict capacitance variation in various surroundings and target shape conditions.
- Contributed to DM37/08 compliance of the test rig contract specification, including power and signaling cabling selection and proper installation.

SNC1-2 steam turbine performance evaluation in 6.2 MW cogeneration plant

[GE Oil & Gas](#)

📅 09/2015 - 04/2016

📍 Florence, Italy

- Defined novel pressure hookups method at lower cost and better performance to achieve accuracy controlled inter-stage enthalpy measurements according to ASME PTC 19.11.
- Programmed a CAN to Modbus TCP embedded module to enable static torque measurements on a MagCanica telemetry.



External focus | GE (06/2014)

Bronze award for the RCA activities on Cessao Onerosa, ADRE monitoring system EMI noise mitigation and customer assurance



Giorgio Barzilai Prize | SIEm (09/2012)

Best paper award for young researchers, for the work "Model order reduction in Finite Element analysis of phased array antennas" presented at the XIX RiNEm



Ph.D. studies grant | UniFi (12/2010)

Full tuition fees and allowance for research on Electromagnetic Compatibility modeling techniques in variable speed drive systems - with GE/Nuovo Pignone



Dean's listing | UniFi (04/2010)

Commissione di Laurea Specialistica in Ingegneria Elettronica presieduta dal Prof. Piero Tortoli (Prot. 1051 n. class. III/9.1.2 del 24 Maggio 2010)



LLP/Erasmus Placement grant | UniFi (10/2009)

Thesis work semester mobility support at Universität des Saarlandes

CERTIFICATIONS



Professional Engineer, ID:ES2010277049000038

Università degli Studi di Firenze

COURSES

Command of the Message

[Ansys](#)

📅 06/2020

📍 Remote

JAWS: Just Another Way of Selling

[Ansys](#)

📅 02/2020

📍 Paris, France

HBM Dynamic Load Cells

[GE Oil & Gas](#)

📅 05/2016

📍 Bologna, Italy

GE Crotonville: Delivering Customer Impact

[GE Oil & Gas](#)

📅 03/2016

📍 Munich, Germany

GE Control Valves — Advanced

[GE Oil & Gas](#)

📅 09/2015

📍 Florence, Italy

CE Marking Overview

[GE Oil & Gas](#)

📅 11/2014

📍 Florence, Italy

GE-6G PowerGen Module Structural Vibrations Monitoring System

GE Oil & Gas

11/2015 - 03/2016 Massa, Italy

- Developed ATEX-compliant distributed acquisition system for more than 150 vibration measurement points, with 40 kHz analog bandwidth, WiFi5 backhaul to storage and real-time monitoring, with GPS or daisy-chain clock distribution synchronization
- Hands-on design of 3D assemblies and BOM for timely delivery and cost optimization of certified equipment in less than 4 months (vs typ. 8 months).
- Introduced and consolidated with third party metrology a software based velocity and displacement real-time computation from accelerometers, achieving direct \$150k savings, and subsequent lab processes OPEX benefits.
- Customer value delivered as onsite Turbomachinery Modules structural commissioning with patented solution

NovaLT16 gas turbine, bearings and cooling component tests

GE Oil & Gas

04/2014 - 10/2015 Florence, Italy

- Design and validation of smartbox (field datalogger) for analog signals.
- Defined EMC-aware measurement loops design to achieve noise immunity and enhanced measurement accuracies, rigorously applying EN 61000 guidelines (CENELEC25).

GE Aviation's GE9X High Pressure Compressor

GE Oil & Gas

01/2014 - 04/2014 Massa, Italy

- High density data acquisition system design for GE Aviation's compressor test in Massa.
- Performed mitigation on EMC noise due to 130 kW variable speed ventilation system

BCL306 and BCL317 centrifugal compressors rotor-dynamics

GE Oil & Gas

06/2013 - 03/2014 Florence, Italy

- Performed an Ex-p execution of stability's magnetic exciter, contributed to the resolution of customer's opened non-conformity on rotordynamic vibrations upon conducting an electromagnetic interference on Bently Nevada's 3500 RCA.

Finite Element Software (FES)

Università degli Studi di Firenze

06/2011 - 02/2014 Florence, Italy

Safety Integrity Levels

GE Oil & Gas

10/2014 Florence, Italy

Electrification

GE Oil & Gas

10/2014 Florence, Italy

GE EHS Orientation

GE Oil & Gas

10/2013 Florence, Italy

ATEX Intermediate

GE Oil & Gas

09/2014 Florence, Italy

Total Quality Lean Six Sigma Green Belt

GE Oil & Gas

06/2014 Florence, Italy

PidXp and P&ID

GE Oil & Gas

11/2013 Florence, Italy

Intellectual Property

GE Oil & Gas

11/2013 Florence, Italy

GE Oil & Gas Product Familiarization

GE Oil & Gas

07/2013 Florence, Italy

SKILLS

Engineering Simulations Testing Management

Sales Strategy Leadership Customer Satisfaction

Finite Element Analysis RF Electronics

Semiconductors Scripting Matlab C++ C

Python

LANGUAGES

Dutch

English

French

German

- FES is an open source framework for the development of “ad-hoc” simulation software aimed at a rapid implementation of novel numerical methods based on the Galerkin method. Core formulations verified reusing meshes from commercial packages (Comsol, HFSS)

Phased array antennas model order reduction

Universität des Saarlandes

📅 10/2009 - 02/2010

📍 Saarbrücken, Germany

- Developed in Matlab efficient near-to-far-fields formulations for fast electromagnetic radiation computation in large finite elements problems exploiting model order reduction with spectral interpolation for beam steering and pattern angles
- Modeled a large patch antenna array, assessing accuracy and performances through mesh/model export from Ansoft HFSS v11

RS-232 modem in ISM 868 MHz band

University of Florence

📅 01/2007 - 04/2007

📍 Florence, Italy

- Designed and prototyped an 868 MHz wireless link for a ground penetration radar remote control over RS-232 protocol
- C-coded the PIC microcontroller FW and optimized in Assembly to guarantee the full-duplex operation over half-duplex TI's CC1100 radio.
- Evaluated the performance of the wireless link in open field

African Great Lakes legacy

📅 08/2026 -

📍 Florence, Italy

- From Bunyoro-Kitara Empire to modern days leveraging digital content creation.

Italian



Kirundi



Spanish



PUBLICATIONS

📄 Patents

- L. Ntibarikure, “Module with sensors arrangement,” US2018018042018. [Online]. Available: <https://patents.google.com/patent/US20180180464A1/en>

📖 Books

- L. Ntibarikure, *Contributions to the Art of Finite Elements in Electromagnetics*. Florence, Italy, 2014. [Online]. Available: <https://hdl.handle.net/2158/843133>

📄 Journal Articles

- L. Ntibarikure, “Multiphysics simulations for 5g rfics and socs,” *Bits&Chips*, 2019.
- F. Barone, A. Signorini, L. Ntibarikure, T. Fiore, F. Di Pasquale, and C. J. Oton, “Fiber-optic liquid level sensing by temperature profiling with an fbg array,” *Sensors*, vol. 18, no. 8, 2018. [Online]. Available: <https://www.mdpi.com/1424-8220/18/8/2422>
- L. Ntibarikure, G. Pelosi, and S. S. and, “Harmonic balance domain decomposition finite elements for nonlinear passive microwave devices analysis,” *Electromagnetics*, vol. 34, no. 3-4, pp. 239–252, 2014. DOI: 10.1080/02726343.2014.877756
- L. Ntibarikure, G. Pelosi, and S. Selleri, “Efficient harmonic balance analysis of waveguide devices with nonlinear dielectrics,” *IEEE Microwave and Wireless Components Letters*, vol. 22, no. 5, pp. 221–223, 2012. DOI: 10.1109/LMWC.2012.2192420

👤 Conference Proceedings

- L. Ntibarikure, G. Pelosi, and S. Selleri, “Assessment of the performances of gmres(r) using a domain decomposition approach as a preconditioner,” in *Proceedings, XX Riunione Nazionale di Elettromagnetismo (XX RiNEm)*, Padua, Italy, 2014. [Online]. Available: <https://inspirehep.net/literature/1403858>
- L. Ntibarikure, “Model order reduction in finite elements analysis of phased array antennas,” in *Proceedings, XIX Riunione Nazionale di Elettromagnetismo (XIX RiNEm)*, S. d. E. SiEm, Ed., Roma, Italy, 2012.
- L. Ntibarikure, G. Pelosi, and S. Selleri, “Harmonic balance finite element analysis of third order intermodulation products in ferrite devices,” in *Proceedings, XIX Riunione Nazionale di Elettromagnetismo (XIX RiNEm)*, S. d. E. SiEm, Ed., Roma, Italy, 2012.